

Actividad | #2 | Calculando Direcciones

Administración de Redes y Servicios.

Ingeniería en Desarrollo de Software



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Desarrollo.
Comandos utilizados

Cálculo de direcciones para VLAN de GERENCIA (192.168.10.0)

Concepto	Calculo	Resultado
Dirección de red.	192.168.10.0	192.168.10.0
Máscara de subred.	/24 → 255.255.255.0	255.255.255.0
Primera IP utilizable.	Red + 1	192.168.10.1
Ultima IP utilizable.	Broadcast – 1	192.168.10.254
Dirección de broadcast.	192.168.10.255	192.168.10.255
Puerta de enlace.	Asignada en la actividad	192.168.10.254

Asignación de IPs para PCs de GERENCIA (VLAN 10):

PC	Dirección IP	Mascara	Gateway
PC1	192.168.10.1	255.255.255.0	192.168.10.254
PC2	192.168.10.2	255.255.255.0	192.168.10.254

Cálculo para VLAN de OPERATIVOS (192.168.20.0)

Concepto	Calculo	Resultado
Dirección de red.	192.168.20.0	192.168.20.0
Máscara de subred.	/24 → 255.255.255.0	255.255.255.0
Primera IP utilizable.	Red + 1	192.168.20.1
Ultima IP utilizable.	Broadcast – 1	192.168.20.254
Dirección de broadcast.	192.168.20.255	192.168.20.255
Puerta de enlace.	Asignada en la actividad	192.168.20.254

PC	Dirección IP	Mascara	Gateway
PC1	192.168.20.1	255.255.255.0	192.168.20.254
PC2	192.168.20.2	255.255.255.0	192.168.20.254

Capturas de pantalla.

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.254

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::200:CFF:FEE6:62DD

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.254

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:F3FF:FEE5:8952

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.1

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.254

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20C:CFFF:FEDC:C6BE

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.254

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::230:A3FF:FE50:601B

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ip config
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::200:CFF:FEE6:62DD
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.10.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>
```

☐ Top



